Acute Pulmonary Edema

Respiratory Assessment
Consider other causes such as pneumonia and ARDS

Consider ALS Back-Up

Position the Patient
In seated position, with head and torso elevated as much as possible.
Feet and legs dependent

Titrate Oxygen
to maintain Sp02 >94%
EMR

12-Lead ECG
STEMI?
N-STEMI?
Cardiac Chest Pain?
Go to Chest Pain CPG

GTN
If SBP >100mg
0.4mg SL every 5 minutes
(maximum 3 doses)
EMT

Consider Salbutamol
5mg Neb
EMT

Consider Tele EMS

Flowsafe CPAP
EMT-A

CPAP / BiPAP
P

Consider Furosemide
40mg IV
P

Reassess

Patient Care Goals

1. Stabilize patients presenting with pulmonary edema.
2. Enhance oxygenation and ventilation.
3. Provide CPAP to moderate and severe cases.
4. Identify and address underlying causes.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients with a provisional diagnosis of pulmonary edema.

Exclusion Criteria

- Patients with difficulty of breathing secondary to another cause e.g. asthma/COPD.

Key Patient Assessment Points

1. **Respiratory Assessment**
   - Common signs and symptoms of pulmonary edema:
     - Difficulty breathing on exertion or when lying down.
     - Wheezing and coughing.
     - Pink sputum/haemoptysis.
     - Chest pain.
     - Cyanosis/pallor/anxiety.
     - Tachycardia.
     - Hypotension.
     - Crackles on auscultation.
2. **Assessment and Determination of Severity**
   - Assess; general appearance, respiratory rate, work of breathing, words per breath, air movement (tidal volume), SpO2.
   - Determine severity as mild, moderate or severe.
3. **Consider alternative causes**
   - Potential causes: Pneumonia, ARDS.
4. **Cardiac Assessment and 12-Lead ECG**
   - Assess for chest pain and perform a 12-Lead ECG to include or exclude concurrent STEMI or arrhythmia.
5. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Consider ALS Back-Up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be requested for severe cases, or in moderate cases where CPAP is not available.
2. **Position the Patient**
   - In seated position with head and torso elevated as much as possible.
   - Feet and legs dependent.
3. **Administer Oxygen**
   - To maintain SpO2 ≥ 92%.
4. **Consider Tele-EMS Consultation**
   - Consult Tele-EMS if required to confirm CPAP settings. Or/
   - If the diagnosis of pulmonary edema is unclear.
5. **Consider GTN**
   - Indicated for SBP > 100mg.
   - Administer GTN 0.4mg SL, every 5 minutes (maximum 3 doses).
6. **CPAP**
   - CPAP should be initiated in acute pulmonary edema when:

        - SpO2 is less than 90% on room air or
        - The patient is in obvious respiratory distress.
    - Attaining a proper mask fit is key to ensuring effective pressure delivery – most patients will find the CPAP mask uncomfortable or claustrophobic, provide reassurance and coaching to the patient.
    - Consider starting with a lower pressure setting (5cmH_2O) and increase as the patient becomes comfortable with the device.
    - If a good mask seal has been obtained, do not remove the CPAP mask to administer GTN or other oral medications.
7. **BiPAP**
    - Consider BiPAP if inspiratory support is required.
    - Initial Settings:
        - IPAP: 10cmH_20
        - EPAP: 5-10cmH_20
8. **Consider Furosemide**
    - Furosemide should only be administered to patients who exhibit signs of fluid overload.
    - Dose: Administer 40mg IV/IO.
    - Routine administration of furosemide is not recommended.
9. **Consider Salbutamol**
    - If CPAP is available, administer CPAP and administer nebulized salbutamol via the CPAP mask.
    - If no CPAP is available, a single dose of Nebulized salbutamol may be administered.
        - Dose: Administer 5mg Nebulized Salbutamol. Repeat after 10-15 minutes only if improvement in respiratory function is noted.
    - If there is no improvement with nebulized Salbutamol, no further doses should be administered.
10. **If Cardiac Chest Pain, or STEMI**
    - If the patient is complaining of cardiac chest pain or STEMI is found on 12-Lead ECG, provide treatment as per Cardiac Chest Pain CPG.
11. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Monitor for signs of fluid overload before administering furosemide.
- Request for ALS back-up moderate, severe or deteriorating patients.

Clinical Insights & Additional Guidance

- **CPAP & BiPAP**
    - Attaining a proper mask fit is key to ensuring effective pressure delivery – ensure proper mask fit and patient comfort during CPAP/BiPAP use.
    - Many patients will find the CPAP mask uncomfortable or claustrophobic, provide reassurance and coaching to the patient.
    - Consider starting with a lower pressure setting (5cmH_20) and increase as the patient becomes comfortable with the device.
    - Avoid removing CPAP mask for oral medications if a good seal is achieved.

Documentation

- Identification: Document patient demographics and clinical condition.
- Mechanism: Note suspected cause (e.g., pneumonia, ARDS).
- Injuries/Illness: Record signs and symptoms (e.g., cyanosis, crackles).
- Signs/Symptoms: Include vital signs, SpO2, and ECG findings.
- Treatment: Detail oxygen administration, GTN doses, CPAP/BiPAP use, and furosemide or salbutamol administration.
- Allergies: Confirm and document any known allergies.

- Medications: List patient's current medications.
- Background: Provide medical history relevant to pulmonary edema.
- Other: Note ALS back-up request and telemedicine call.

Key Performance Indicators

1. Apply CPAP/BiPAP for all severe pulmonary edema patients.

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Resuscitation